

Appendix: Metadata comparison - output from print.nc on annual and monthly data files

Annual temperature file "tas_hadukgrid_uk_1km_ann_202301-202312.nc"	Monthly temperature file (July) "tas_hadukgrid_uk_1km_mon_202407.nc"
<pre>> print.nc(nc_conn) netcdf netcdf4 { dimensions: time = 1 ; projection_y_coordinate = 1450 ; projection_x_coordinate = 900 ; bnds = 2 ;</pre>	<pre>> nc_conn <- open.nc(here(data_dir_name, data_subdir, data_files_months_temprtr[7L])) > print.nc(nc_conn) netcdf netcdf4 { dimensions: time = 1 ; projection_y_coordinate = 1450 ; projection_x_coordinate = 900 ; bnds = 2 ;</pre>
<pre>variables: NC_DOUBLE tas(projection_x_coordinate, projection_y_coordinate, time) ; NC_DOUBLE tas:_FillValue = 1e+20 ; NC_CHAR tas:standard_name = "air_temperature" ; NC_CHAR tas:long_name = "Mean air temperature" ; NC_CHAR tas:units = "degC" ; NC_CHAR tas:description = "Mean air temperature" ; NC_STRING tas:label_units = "°C" ; NC_CHAR tas:level = "1.5m" ; NC_STRING tas:plot_label = "Mean air temperature at 1.5m (°C)" ;</pre>	<pre>variables: NC_DOUBLE tas(projection_x_coordinate, projection_y_coordinate, time) ; NC_DOUBLE tas:_FillValue = 1e+20 ; NC_INT64 tas:least_significant_digit = 3 ; NC_CHAR tas:standard_name = "air_temperature" ; NC_CHAR tas:long_name = "Mean air temperature" ; NC_CHAR tas:units = "degC" ; NC_CHAR tas:description = "Mean air temperature" ; NC_STRING tas:label_units = "°C" ; NC_CHAR tas:level = "1.5m" ;</pre>

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NC_CHAR tas:cell_methods = "time: mid_range within days time: mean
over days" ;
NC_CHAR tas:grid_mapping = "transverse_mercator" ;
NC_CHAR tas:coordinates = "latitude longitude" ;
NC_INT transverse_mercator ;
NC_CHAR transverse_mercator:grid_mapping_name =
"transverse_mercator" ;
NC_DOUBLE transverse_mercator:longitude_of_prime_meridian = 0 ;
NC_DOUBLE transverse_mercator:semi_major_axis = 6377563.396 ;
NC_DOUBLE transverse_mercator:semi_minor_axis = 6356256.909 ;
NC_DOUBLE transverse_mercator:longitude_of_central_meridian = -2 ;
NC_DOUBLE transverse_mercator:latitude_of_projection_origin = 49 ;
NC_DOUBLE transverse_mercator:false_easting = 4e+05 ;
NC_DOUBLE transverse_mercator:false_northing = -1e+05 ;
NC_DOUBLE transverse_mercator:scale_factor_at_central_meridian =
0.9996012717 ;
NC_INT64 time(time) ;
NC_CHAR time:axis = "T" ;
NC_CHAR time:bounds = "time_bnds" ;
NC_CHAR time:units = "hours since 1800-01-01 00:00:00" ;
NC_CHAR time:standard_name = "time" ;
NC_CHAR time:calendar = "standard" ;
NC_INT64 time_bnds(bnds, time) ;
NC_DOUBLE projection_y_coordinate(projection_y_coordinate) ;
NC_CHAR projection_y_coordinate:axis = "Y" ;
NC_CHAR projection_y_coordinate:bounds =
"projection_y_coordinate_bnds" ;
NC_CHAR projection_y_coordinate:units = "m" ;
NC_CHAR projection_y_coordinate:standard_name =
"projection_y_coordinate" ;
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NC_STRING tas:plot_label = "Mean air temperature at 1.5m (°C)" ;
NC_CHAR tas:cell_methods = "time: mid_range within days time: mean
over days" ;
NC_CHAR tas:grid_mapping = "transverse_mercator" ;
NC_CHAR tas:coordinates = "latitude longitude month_number
season_year" ;
NC_INT transverse_mercator ;
NC_CHAR transverse_mercator:grid_mapping_name =
"transverse_mercator" ;
NC_DOUBLE transverse_mercator:longitude_of_prime_meridian = 0 ;
NC_DOUBLE transverse_mercator:semi_major_axis = 6377563.396 ;
NC_DOUBLE transverse_mercator:semi_minor_axis = 6356256.909 ;
NC_DOUBLE transverse_mercator:longitude_of_central_meridian = -2 ;
NC_DOUBLE transverse_mercator:latitude_of_projection_origin = 49 ;
NC_DOUBLE transverse_mercator:false_easting = 4e+05 ;
NC_DOUBLE transverse_mercator:false_northing = -1e+05 ;
NC_DOUBLE transverse_mercator:scale_factor_at_central_meridian =
0.9996012717 ;
NC_INT64 time(time) ;
NC_CHAR time:axis = "T" ;
NC_CHAR time:bounds = "time_bnds" ;
NC_CHAR time:units = "hours since 1800-01-01 00:00:00" ;
NC_CHAR time:standard_name = "time" ;
NC_CHAR time:calendar = "standard" ;
NC_INT64 time_bnds(bnds, time) ;
NC_DOUBLE projection_y_coordinate(projection_y_coordinate) ;
NC_CHAR projection_y_coordinate:axis = "Y" ;
NC_CHAR projection_y_coordinate:bounds =
"projection_y_coordinate_bnds" ;
NC_CHAR projection_y_coordinate:units = "m" ;
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<pre>NC_DOUBLE projection_y_coordinate_bnds(bnds, projection_y_coordinate) ; NC_DOUBLE projection_x_coordinate(projection_x_coordinate) ; NC_CHAR projection_x_coordinate:axis = "X" ; NC_CHAR projection_x_coordinate:bounds = "projection_x_coordinate_bnds" ; NC_CHAR projection_x_coordinate:units = "m" ; NC_CHAR projection_x_coordinate:standard_name = "projection_x_coordinate" ; NC_DOUBLE projection_x_coordinate_bnds(bnds, projection_x_coordinate) ; NC_DOUBLE latitude(projection_x_coordinate, projection_y_coordinate) ; NC_CHAR latitude:units = "degrees_north" ; NC_CHAR latitude:standard_name = "latitude" ; NC_DOUBLE longitude(projection_x_coordinate, projection_y_coordinate) ; NC_CHAR longitude:units = "degrees_east" ; NC_CHAR longitude:standard_name = "longitude" ;</pre>	<pre>NC_CHAR projection_y_coordinate:standard_name = "projection_y_coordinate" ; NC_DOUBLE projection_y_coordinate_bnds(bnds, projection_y_coordinate) ; NC_DOUBLE projection_x_coordinate(projection_x_coordinate) ; NC_CHAR projection_x_coordinate:axis = "X" ; NC_CHAR projection_x_coordinate:bounds = "projection_x_coordinate_bnds" ; NC_CHAR projection_x_coordinate:units = "m" ; NC_CHAR projection_x_coordinate:standard_name = "projection_x_coordinate" ; NC_DOUBLE projection_x_coordinate_bnds(bnds, projection_x_coordinate) ; NC_DOUBLE latitude(projection_x_coordinate, projection_y_coordinate) ; NC_CHAR latitude:units = "degrees_north" ; NC_CHAR latitude:standard_name = "latitude" ; NC_DOUBLE longitude(projection_x_coordinate, projection_y_coordinate) ; NC_CHAR longitude:units = "degrees_east" ; NC_CHAR longitude:standard_name = "longitude" ; NC_INT64 month_number ; NC_CHAR month_number:units = "1" ; NC_CHAR month_number:long_name = "month_number" ; NC_INT64 season_year ; NC_CHAR season_year:units = "1" ; NC_CHAR season_year:long_name = "season_year" ;</pre>
<pre>// global attributes: NC_CHAR :comment = "Annual resolution gridded climate observations" ; NC_CHAR :creation_date = "2024-05-17T09:50:50" ; NC_CHAR :frequency = "ann" ; NC_CHAR :institution = "Met Office" ;</pre>	<pre>// global attributes: NC_CHAR :comment = "Monthly resolution gridded climate observations" ; NC_CHAR :creation_date = "2024-08-06T10:07:26" ; NC_CHAR :frequency = "mon" ; NC_CHAR :institution = "Met Office" ;</pre>

<pre>NC_CHAR:references = "doi: 10.1002/gdj3.78"; NC_CHAR:short_name = "annual_meantemp"; NC_CHAR:source = "HadUK-Grid_v1.3.0.0"; NC_CHAR:title = "Gridded surface climate observations data for the UK"; NC_CHAR:version = "v20240514"; NC_CHAR:Conventions = "CF-1.7"; }</pre>	<pre>NC_CHAR:references = "doi: 10.1002/gdj3.78"; NC_CHAR:short_name = "monthly_meantemp"; NC_CHAR:source = "HadUK-Grid_Provisional"; NC_CHAR:title = "Gridded surface climate observations data for the UK"; NC_CHAR:version = "provisional"; NC_CHAR:Conventions = "CF-1.7"; }</pre>
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